

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Define cloud computing and Explain any four characteristics that make it different from traditional on-premise computing.

Soln:- Definition:-

cloud computing is a technology that provides computing resources such as servers, storage, database, networking and software through the internet on an as demand basis.

Characteristics:-

1. On demand and Self Service:-

User can access computing resources whenever required without human interaction with service provider.

Broad Network access:- Cloud services can be accessed from anywhere using the internet through computers, mobiles and other devices.

Resource pooling:-

Cloud providers share computing and storage among multiple users using virtualization technology.

Resource Elasticity:-

Resource can be increased or decreased quickly based on user requirements.

Difference from traditional computing:-

Traditional computing requires physical hardware and the maintenance while cloud computing provides flexible resources over the internet.

Q2 Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Evolution of cloud computing

Hardware Evolution

Early system used large mainframe computers.

Personal computers and powerful servers computing capabilities.

Virtualization allowed multiple virtual machines on one physical sharing.

Distributed system:-

* computing resources were connected across multiple systems

Technologies like grid computing and cluster computing improved resource sharing.

Internet Evolution

* Growth of high speed internet enabled remote access to computing resources.

Web technologies allowed applications to be delivered online.

Internet Software Evolution

Web Applications, APIs, and service-oriented architecture Enabled online services.

How It Enabled modern cloud platforms:-

- * These developments created scaled platforms like public, private and hybrid clouds.
- * Services such as IaaS, PaaS, and SaaS became possible.

Q3

Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Ans:-

Role of server virtualization

- * Server virtualization divides one physical server into multiple virtual machines.
- * It improves resource utilization and reduces hardware cost.
- * It provides flexibility, scalability and efficient management of cloud resources.

Difference

Virtualization

It is a technology that creates virtual resources

Uses Hypervisors to create virtual machines

Focus on resource management

Ex:- VMware, Hyper-V

cloud computing

It is a service model that delivers resources through the Internet

Provides services like IaaS, SaaS, PaaS

Focuses on service delivery to user

Ex:- AWS, Microsoft Azure.

Q4 Discuss the role of data centers in delivering cloud services

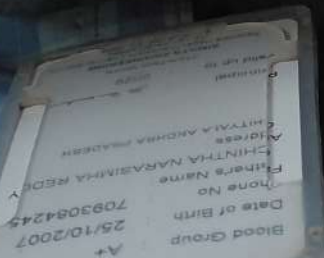
Role of Data centers:-

1. Resource Hosting

* Data centers contain servers, storage devices and networking equipment required for cloud services

Data Storage:-

* The data storage and large amounts of user data securely.



Service Delivery:-

They provide applications, database, and computing resources through the Internet

High Availability:-

Redundant system and redundancy ensure continuous services availability.

Security management:-

Data centers use Firewall, Encryption and monitoring system to protect data.

85 Differentiate IaaS, PaaS, SaaS and XaaS with one suitable real-world example for each model

Model	Description	Provider Responsibility	User Responsibility	Ex:
IaaS	Provides Virtual Infrastructure like server, storage and networks	Hardware, Storage, Networking	OS, Application, data	Amazon EC2
PaaS	Provides a platform and deploying	Infrastructure, runtime tools	Application development	Google App
SaaS	Provides ready to use software application over the Internet	Application, Updates, Security	Using Software	Gmail, Microsoft
XaaS	Provides any IT services through the cloud	Complete Service management	Using required services	Salesforce, cloud service

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow